DATA SCIENCE

If I Had a Data Science Interview in 24 Hours, I'd Study Only This

An hour-by-hour plan for the last 24 hours, with the exact topics, answers and code to revise. Nothing extra.

INSIDE THIS PDF · 8 PAGES

- The 24-hour plan
- Hours 1-4: Statistics
- Hours 5-9: Machine learning
- Hours 10-13: SQL and Python
- Hours 14-16: Case question
- Hours 17-19: Your projects
- Last-night checklist

@thedataguy16

The 24-hour plan

Study in this order. The most asked topics come first, so even if you stop early you have covered what matters.

- **Hours 1-4: Statistics**
  p-value, tests, A/B testing, distributions. Asked in almost every round.
- **Hours 5-9: Machine learning**
  Core algorithms, metrics, overfitting, imbalance.
- **Hours 10-13: SQL and Python**
  Window functions and Pandas groupby. Usually a live coding round.
- **Hours 14-16: Case question**
  Metric dropped or design an experiment.
- **Hours 17-19: Your own projects**
  The round most people lose.
- **Hours 20-21: One mock interview**
  Say answers out loud. Record yourself.
- **Hours 22-24: Sleep**
  A tired brain forgets definitions. Do not skip this.

@thedataguy16

Hours 1-4: Statistics

Know these answers in one or two lines each.

- **p-value**

  If nothing really changed, the chance of seeing a result this extreme. Below 0.05 usually counts as significant.

- **Type I vs Type II error**

  Type I is a false alarm. Type II is missing a real effect, usually from a small sample.

- **Central Limit Theorem**

  Averages of large samples are roughly normal, whatever the original data looks like. That is why t-tests work.

- **Which test**

  Two means: t-test. Three or more: ANOVA. Two categories: chi-square. Two rates: z-test.

- **A/B test steps**

  One metric, fix sample size first, random split, run full weeks, no peeking, check guardrails.

@thedataguy16

Hours 5-9: Machine learning

Interviewers test understanding, not memory. Be ready to explain trade-offs.

Bias vs variance

High bias: too simple, underfits. High variance: memorises, overfits.

Random Forest vs Gradient Boosting

Forest builds trees in parallel and averages. Boosting builds them one after another, each fixing the last one's errors.

Precision vs recall

Precision: of what I flagged, how much was right.
Recall: of all real cases, how many I caught.

Imbalanced data

Do not trust accuracy. Use recall, F1, PR-AUC. Fix with class weights or resampling.

Regularisation

L1 (Lasso) can drop features to zero. L2 (Ridge) shrinks all of them.

Data leakage

A feature that would not exist at prediction time. Gives great test scores and fails in production.

@thedataguy16

Hours 10-13: SQL and Python

Practise these exact patterns. Most coding rounds are a variation.

```
-- Top 3 products by revenue per category
SELECT * FROM (
  SELECT category, product, SUM(revenue) rev,
    RANK() OVER (PARTITION BY category
      ORDER BY SUM(revenue) DESC) rk
  FROM sales GROUP BY category, product) t
WHERE rk <= 3;

# Same in Pandas
s = df.groupby(["category","product"])
["revenue"].sum()
s.groupby(level=0, group_keys=False).nlargest(3)
```

Also revise

LAG for period change, LEFT JOIN with IS NULL, merge and pivot_table in Pandas.

@thedataguy16

Hours 14-16: Case question

"Daily orders dropped 15% yesterday. What do you do?" Use this structure for any metric question.

1. Check the data first

Is tracking broken? Did a dashboard or definition change?

2. Internal or external

Release, pricing change, outage? Or holiday, competitor, weather?

3. Slice the drop

By platform, city, new vs returning users, funnel step. Find where it is concentrated.

4. Form a hypothesis and test it

"Drop is only on Android after yesterday's app update."

5. Recommend

Roll back, fix, or monitor. Say what metric would confirm it.

@thedataguy16

Hours 17-19: Your projects

Prepare a 2-minute story for each project on your resume. Use this order.

- **Problem**
  What business question, for whom, why it mattered.
- **Data**
  Source, size, the messiest cleaning issue you fixed.
- **Approach**
  Baseline first, then the model you chose and why not the alternatives.
- **Result in numbers**
  "Recall went from 0.52 to 0.81" or "saves ₹12 lakh a year".
- **What you would do differently**
  Shows maturity. Every interviewer asks it.

Rule: if you cannot explain a line on your resume for 2 minutes, remove it before the interview.

@thedataguy16

Last-night checklist

Hours 20-24. Do these, then stop studying.

- **One mock interview**

  Ask a friend or record yourself answering 5 questions from this PDF.

- **3 questions to ask them**

  Team's biggest data problem, how success is measured, what the first 90 days look like.

- **Setup ready**

  Laptop charged, IDE or SQL editor open, resume printed or on screen.

- **Sleep 6+ hours**

  Recall drops sharply when you are tired. The last 3 hours of study are worth less than sleep.

@thedataguy16